PRS PRACTICE MODULE · FINAL PRESENTATION

The Eyes

Multi-view perception and scene understanding for robot manipulation

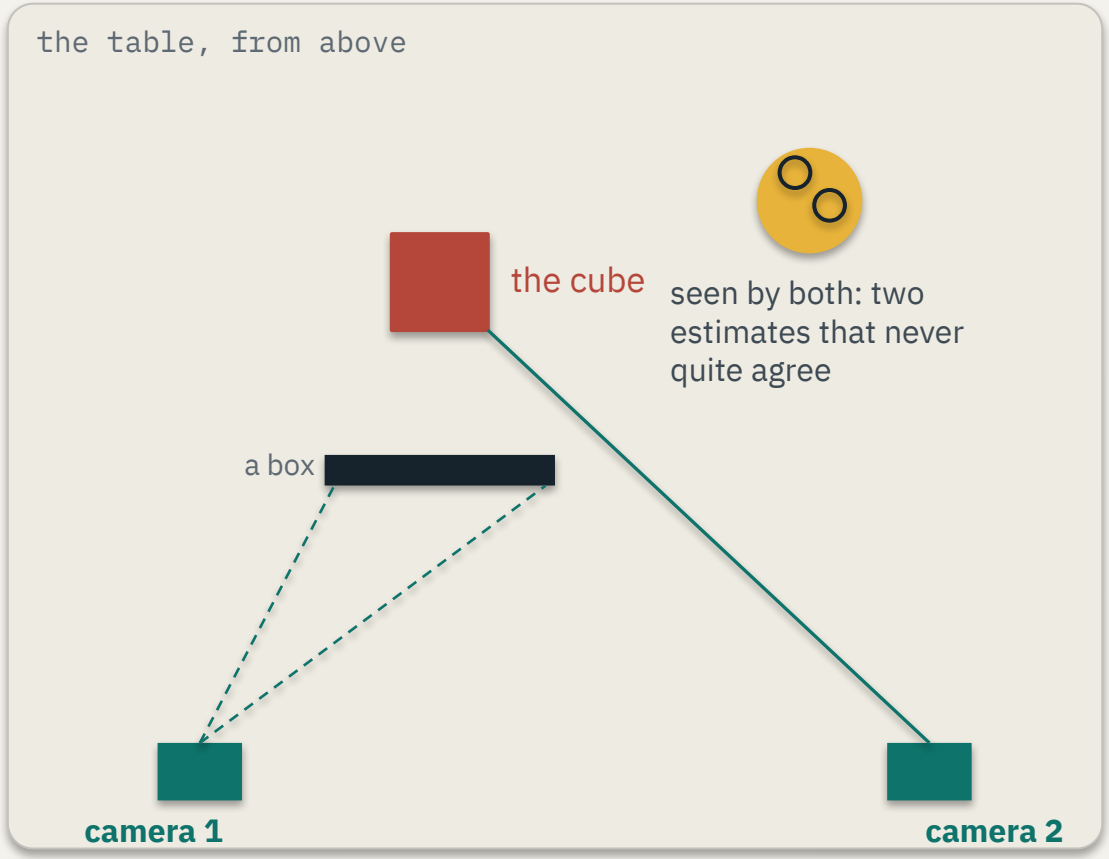
Group 12 · AIS08 · Chan Wai Jye (solo)
31 October 2026

Code built with Claude Code under the author's design and review



The rig: two RealSense D435i on tripods, the printed objects in their marker trays

Why one camera is not enough



A planner works with statements, like `(on cube table)`. A real robot has to get them from cameras.

1 Association

Which estimates, one per camera, are the same object?

2 Arbitration

When two views disagree, which one is wrong?

3 Symbolisation

How do poses become facts reliable enough to plan with?

The rig



2 × Intel RealSense D435i

Active IR stereo depth plus colour, on tripods, each selected by its serial number

About 0.74 m apart, 80° between views

Frames paired by timestamp, never more than 16.7 ms apart

Calibrated to about 0.3 px

Against the ChArUco board, and registered again at the start of every session

Three printed objects

Printed from the pose estimator's own meshes; the grey marker trays are for ground truth only

Source: config/extrinsics.yaml; frames.skew_ms of every scored scene and take

One C++ process, two views

PER VIEW · once for each camera

2 × D435i
RGB-D frames, 30 a second

→

RGB-D YOLOv8n
4-channel input; trained here on 40,012 synthetic frames, fine-tuned on 360 real

→

FoundationPose
6-DoF pose from the CAD mesh; pretrained, frozen

→

World frame
Each pose moved into the shared table frame

PER FRAME PAIR · the project's core, written from first principles in C++

Depth check ★
Does the other camera see through this pose?

→

Association ★
The same object, matched across the views

→

Fusion ★
Gate at 50 mm / 30°; confidence weighting; an EKF per object

→

Scene graph ★
on, in, clear, occluded_from; the support check

→

Outputs
PDDL (:init ...) facts and ObjectState

★ No sensor-fusion, pose-averaging or scene-graph library: Eigen for linear algebra only. The code was written with Claude Code under the author's design and review.

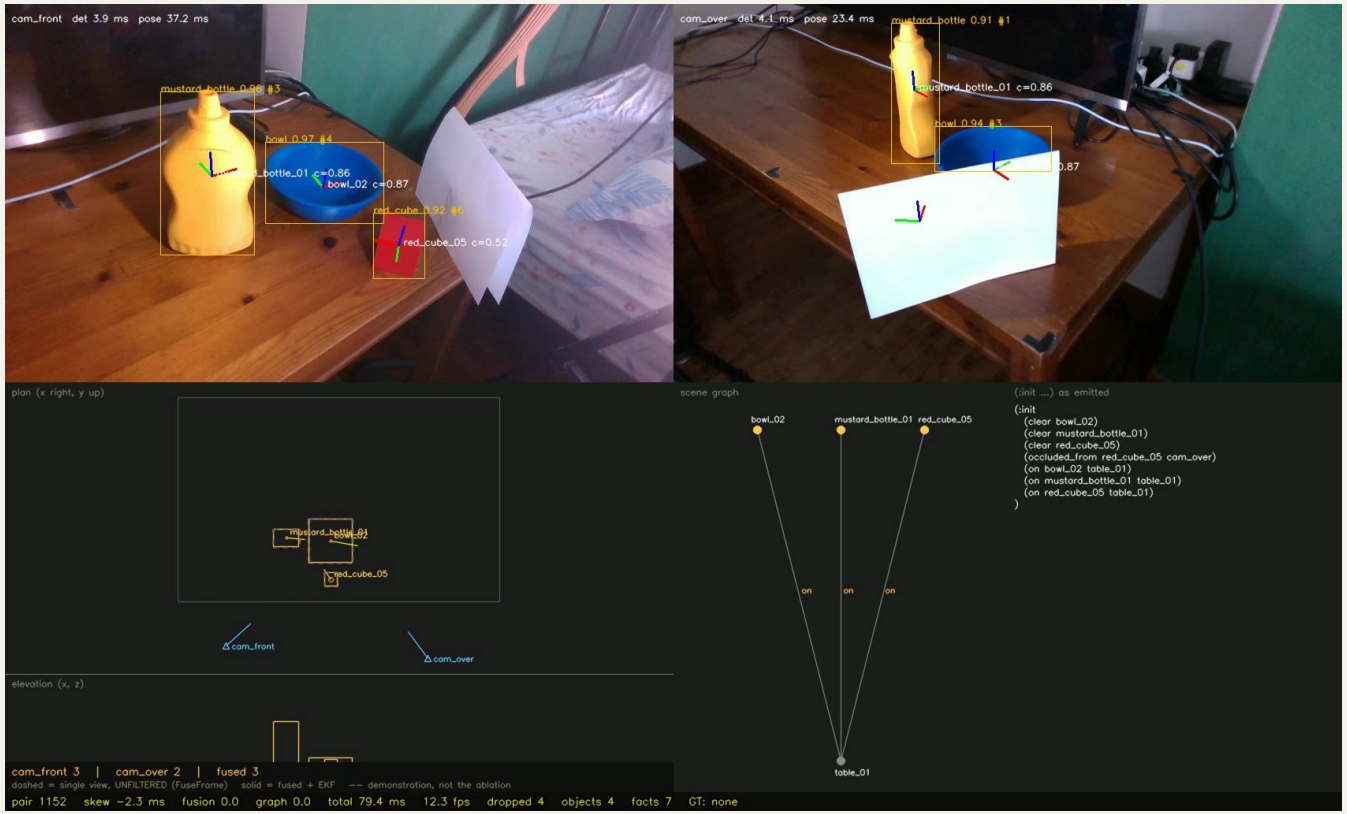
What the console shows



Source: the live take demo_live_d, 25 Oct 2026, at 60 s: the cube is in the bowl

1. **Both camera views**
   Boxes from the detector, axes from the pose estimator, per camera
2. **The table from above**
   The fused objects in the shared frame, and the two cameras
3. **The scene graph**
   Each object and what it rests on or sits in
4. **The facts**
   The (:init …) block a planner reads

Live on the rig



LIVE, one take, not replayed. At 84 s the card hides the cube from cam_over: cam_front 3, cam_over 2, fused 3.

13.3
frame pairs a second over the 195 s take

10
a second with three objects tracked in each view

79%
of a three-object pair is the pose estimator

1.1 s
per new object per view, while the pose estimator registers it

Source: demo_live_d, 25 Oct 2026, its metrics.sqlite; final report §9.8

Ground truth from printed marker trays



A ground-truth still: the cube in its tray, with the tray's markers



The pose the markers give, drawn back onto the frame (green)

Within 1.5 mm
The two cameras agree on each tray's true pose

Recorded once
300 frame pairs a scene, replayed for every method

Under 10% of the diameter
ADD-S, the surface-to-surface error, counts as right: 8.7 mm for the cube

Source: final report §7–8; ground truth across cameras 0.3–1.4 mm (docs/OCTOBER_PLAN.md); config/meshes.yaml

Sixteen scenes, four levels of occlusion

T1

Alone

5 scenes

A cube or the bottle, nothing else

T2

In clutter

3 scenes

Three objects among distractors

T3

Half hidden

4 scenes

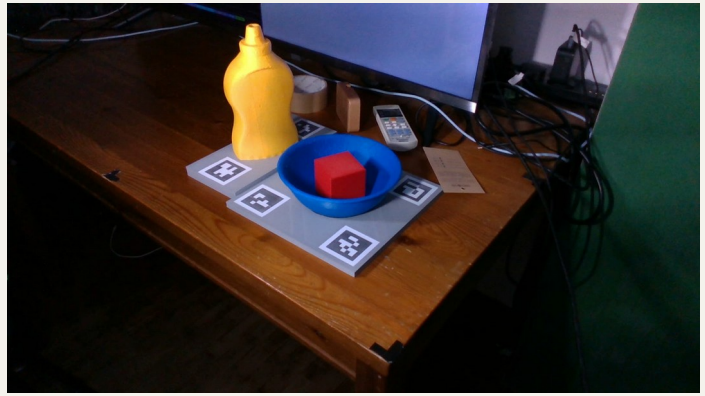
From one camera; one scene's level not certified

T4

Fully hidden

4 scenes

From one camera, which sees 0–6% of it



Two more with the cube inside the bowl

`(in red_cube bowl)` is reported on 300 of 300 frame pairs in both.

Every method runs on the same recorded frames. Intervals are 95% bootstrap over scenes; a difference counts only if it is beyond the 2–4 mm sensor floor.

Source: final report §8, §9.4; t2_in_a and t2_in_b, 10 Oct

A hidden object is still found



t4_cube_over: cam_front (left) sees the cube; from cam_over (right) the card hides it

Scene	Blocked camera sees	It finds it on	Fused finds it on	Fused ADD-S
t4_cube_over	0%	0 / 300	**300 / 300**	3.08 mm
t4_cube_front	0%	0 / 300	**300 / 300**	1.65 mm
t4_bottle_over	6%	0 / 300	**300 / 300**	3.91 mm
t4_bottle_over_b	2%	0 / 300	**300 / 300**	2.85 mm

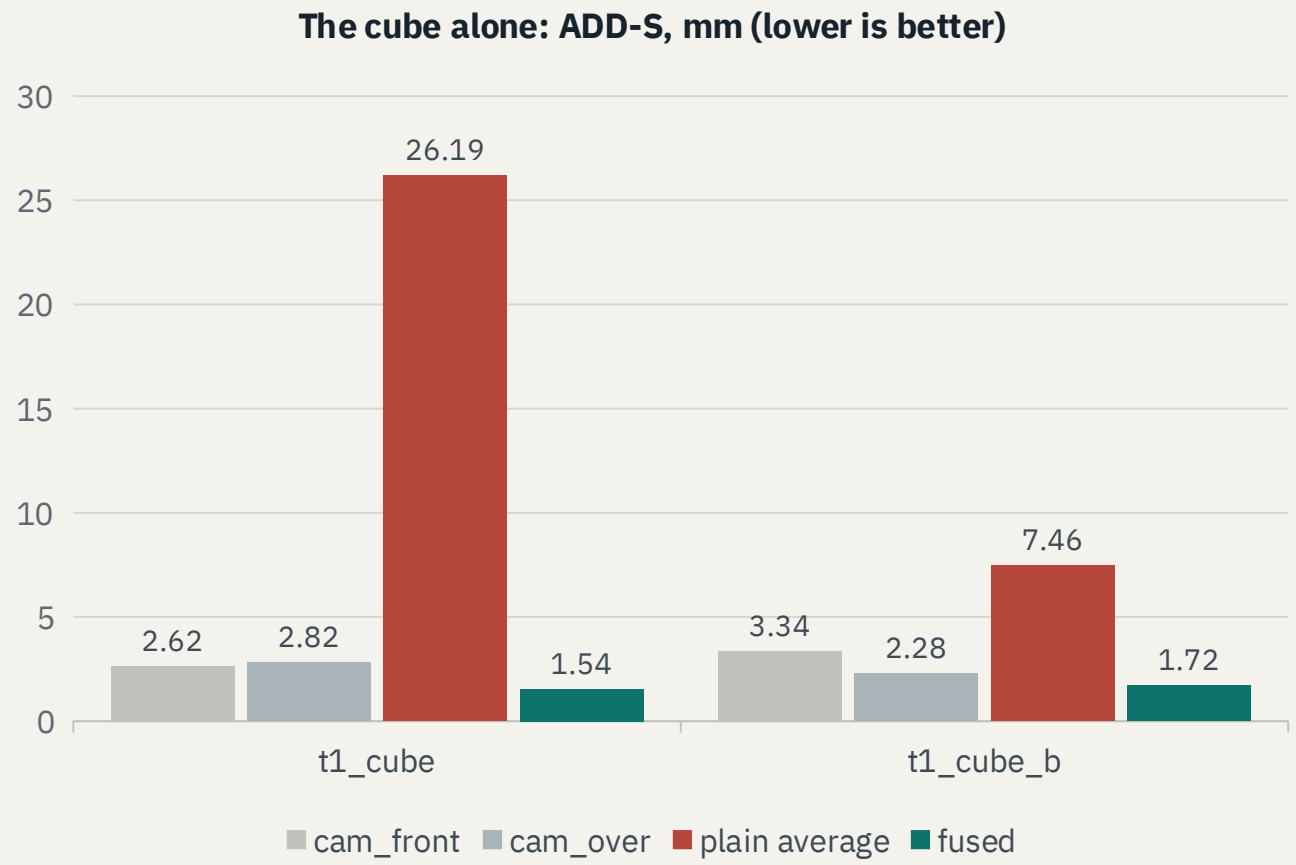
Source: final report §9.4; the share each blocked camera sees from runs/scored/*/visibility.csv

+1.00

recall, fused against the blocked camera, 95% interval [1.00, 1.00]

A single camera does as well only when it is the open one. No single camera covers all four scenes.

How the views are combined matters



Averaging fails

The cube looks the same from several directions; the cameras pick different ones, and their average is a rotation neither saw.

Gated vs plain average

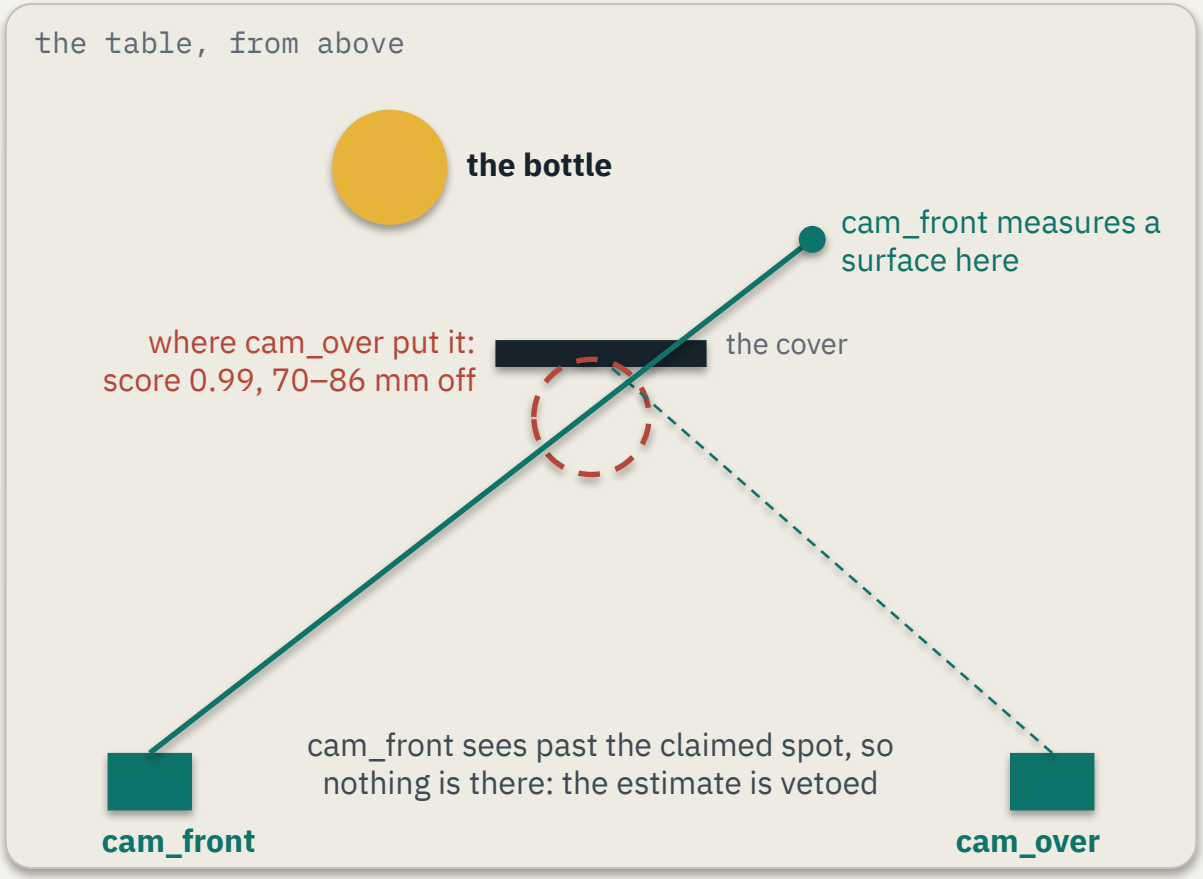
All 16 scenes: −9.81 mm [−16.96, −3.75], beyond the sensor floor.

Fused vs the better camera

All 16: 2.76 against 3.19 mm, a difference of −0.36 [−0.83, 0.08]. Not resolvable on this rig.

Source: runs/scored/20261005/t1_cube*/ablation.log (all arms with the EKF); final report §9.4, docs/results/ablation_ci.json

When fusion believed the wrong camera



Keep the higher score

70.75 and **83.72 mm**

the good camera alone: 3.06 and 3.14 mm

Veto by the other camera's depth

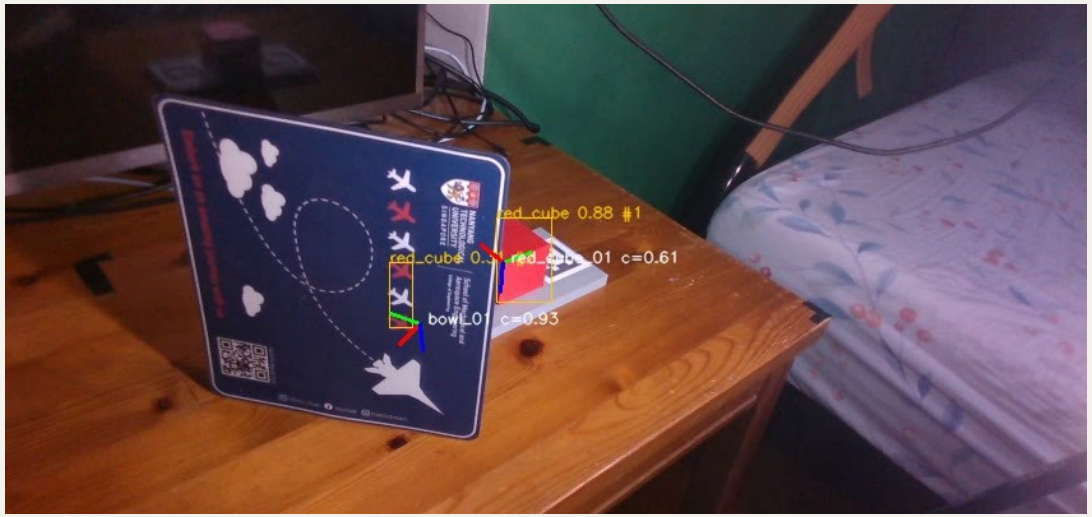
3.03 and **3.12 mm**

1,035 estimates vetoed

Over all 22 recordings. One of them was within 30 mm of the truth, of 17,121 that were.

Source: final report §9.4–9.5; t3_bottle_over_b and t3_bottle_over; refusals over all 22 scenes, docs/OCTOBER_PLAN.md, 17 Oct

Objects that are not there



Synthetic-only detector, cam_front: a cube boxed on the card's red print



Fine-tuned detector, the same frame: no false object

The support check: 2,771 → 1

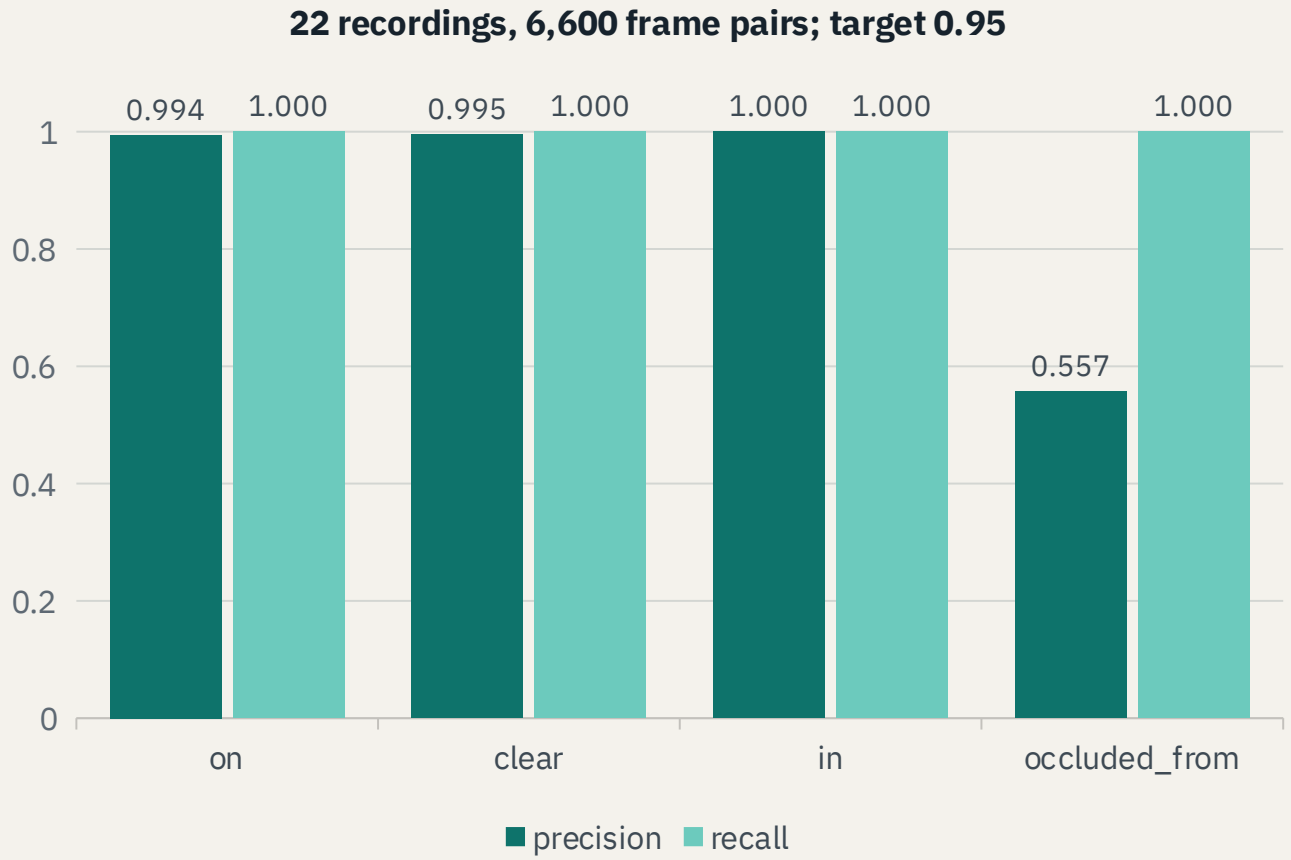
An object is reported only if something supports it. Seven scenes: false object-pairs 2,771 → 1, all 2,100 real ones kept.

Fine-tuning: 731 → 0

On 360 real frames of ten unscored layouts. mAP@0.5:0.95 0.670 → 0.934; false objects over 22 recordings 731 → 0.

Source: final report §9.2, §9.7; replays of 20261005/t3_cube_front with each detector

Facts checked against hand labels



Source: final report §9.7; scripts/predicate_accuracy.py, docs/predicate_labels.yaml

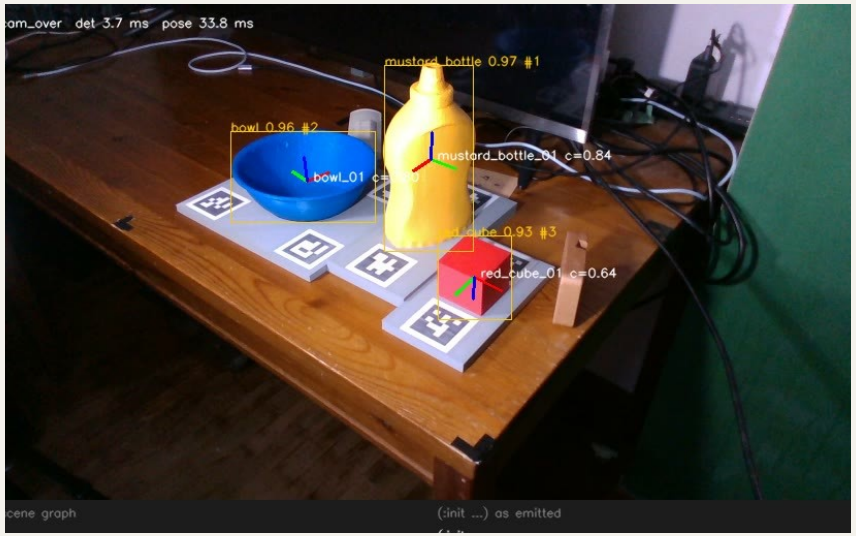
on, in, clear: met

Precision 0.994 to 1.000, recall 1.000. The few false on and clear facts are one duplicate bottle in one dim scene.

occluded_from: 0.557

A half-hidden object (44–68% visible) reads as hidden from the camera whose estimate was refused. The definition fixed in advance is under 25% visible. And the fact cannot tell a blocked camera from a detector that missed.

What it still gets wrong



t2_clutter_c from cam_over: the cube is small and far away. Fused, it is inside the threshold on 75.7% of frames

occluded_from

A half-hidden object reads as hidden; precision 0.557

A hand in view

A pose can register on the hand; a passing hand can start a duplicate track for one frame

Registration stalls

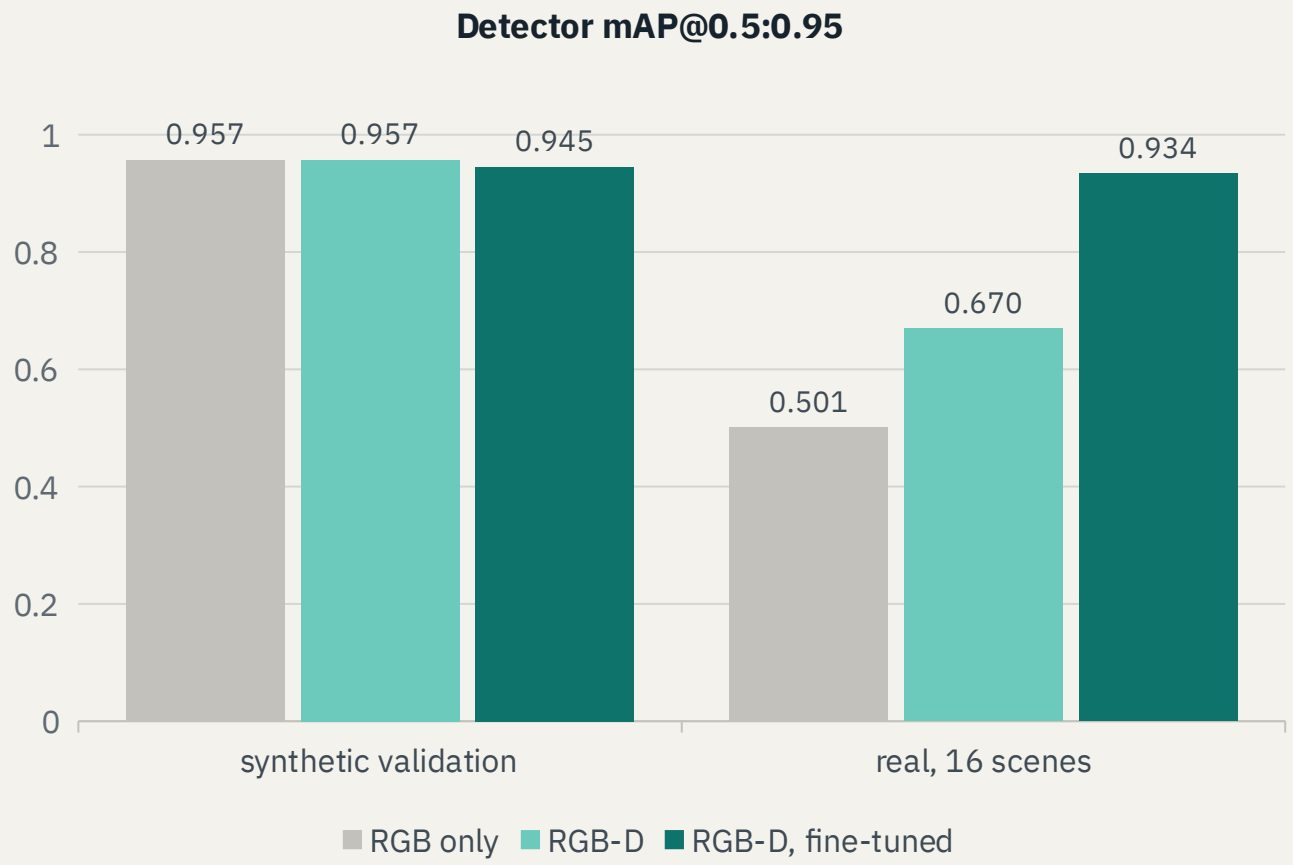
About 1.1 s per new object per view, with nothing processed meanwhile

One rig, one room

Still scenes for every accuracy figure; four distractors, not eight

Source: final report §9.4, §9.7, §9.8, §11; the live takes of 25 Oct

On real frames, depth is what finds the cube



Source: final report §9.2, §9.9; 9,988 synthetic validation frames, 960 real frames of the 16 scored scenes

+0.00076
what depth added on synthetic frames: nothing

+0.169
what it adds on real frames; the cube's AP@0.5 goes 0.099 → 0.905

0.934
after fine-tuning on 360 real frames: most of the sim-to-real gap closed

Four kinds of technique in one system

1 Supervised deep learning

An RGB-D YOLOv8n trained on 40,012 synthetic frames from Isaac Sim, then fine-tuned on 360 labelled real ones

2 A multimodal network

A 4-channel early-fusion stem, its depth channel made by inflating the COCO weights; FoundationPose, a pretrained transformer; ONNX Runtime on CUDA

3 Hybrid estimation

Two per-view estimators combined by confidence weighting, a disagreement gate and depth evidence; a learned front end with a classical EKF behind it

4 Intelligent sensing

Two calibrated RGB-D cameras; each camera's depth used as evidence against the other's poses; poses turned into relational facts

Source: final report §4–7; CLAUDE.md, technique map

What a planner receives

(:init …) as emitted, live, at 60 s

```
(:init
  (clear bowl_02)
  (clear mustard_bottle_01)
  (clear red_cube_04)
  (in red_cube_04 bowl_02)
  (on bowl_02 table_01)
  (on mustard_bottle_01 table_01)
)
```

Source: demo_live_d at 60 s; INTERFACES.md §5

ObjectState: a fixed contract

`id, cls` instance and class

`pose_w` fused 6-DoF pose, table frame

`confidence` fused score

`relations` filled by the scene graph

`visible_views, observed_views` fused from, seen by

`extent` half-extents, axis-aligned

`position_sigma_m, orientation_sigma_rad` uncertainty

`velocity_w, stamp_ns` motion and time

Fields may be added, never removed or re-typed: the capstone, MANIP-WM, builds on it.

THE EYES · GROUP 12

Two cameras do see more than one

But the second camera matters most as a witness: it is what tells you when the first one is wrong.

2.76 mm

fused ADD-S over the 16 scored scenes

every frame

an object hidden from one camera, still found

9.8 mm

the gate's margin over a plain average

≥ 0.994

precision of on, in and clear

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